46770 Integrated Energy Grids

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Lecture 3 – Capacity and dispatch optimisation in one node

Types of optimisation problems and course structure

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One-node dispatch optimisation
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Joint capacity and dispatch optimisation in one node
(Lecture 3)</td><td colspan="4">Joint capacity and dispatch optimisation in a network
(Lecture 10)</td></tr>
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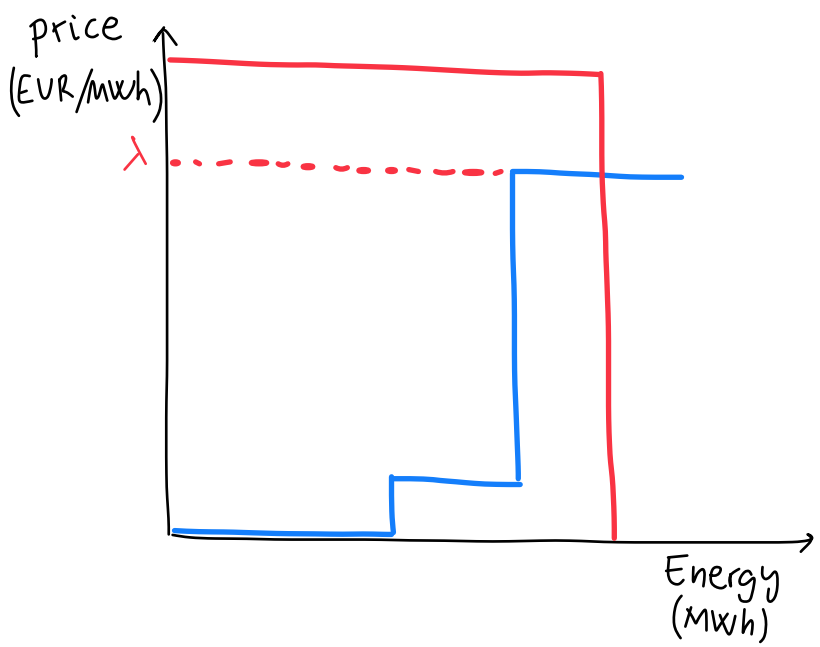
Review of Lecture 2

Economic dispatch example

	Wind	Solar	Gas
Variable cost o_s (EUR/MWh)	0	5	50
Installed Capacity G_s (MW)	2	1	1
CF_s	0.5	0.5	1
Calculate generation g_s (assuming demand d = 1.5 MWh)	1	0.5	0
Calculate generation g_s (assuming demand d = 2 MWh)	1	0.5	0.5



Economic dispatch is also called merit-order dispatch (because we rank the generators based on their merit)

Economic dispatch is used to decide which generators produce energy in a market.

Economic dispatch or one-node dispatch optimisation (II)

Assume we have a set of generators s (e.g., onshore wind, solar PV, gas power plant…) each of them has an installed capacity G_s and a linear variable cost o_S. The economic dispatch consists in calculating the optimal dispatch (how much energy is being produced by each generator g_s) to supply the demand d in a certain hour while minimising the total system cost.

$$\min_{g_s} \sum_s o_s g_s$$

subject to:

$$\sum_s g_s - d = 0 \leftrightarrow \lambda$$

$$-g_s + G_s \geq 0 \leftrightarrow \overline{\mu_s}$$

To find a solution, we start by building the Lagrangian function

$$\mathcal{L}(g_s, \lambda, \overline{\mu_s}) = \sum_s o_s g_s - \lambda(\sum_s g_s - d) - \sum_s \overline{\mu_s}(-g_s + G_s)$$

We derive the Lagrangian and set the derivative equal to zero

$$\frac{\partial \mathcal{L}}{\partial g_s} = o_s - \lambda + \overline{\mu_s} = 0.$$

In the optimum (*), the inequality constraint can be binding ($\overline{\mu_s^*} > 0$) when the installed capacity is limiting the generation or not-binding ($\overline{\mu_s^*} = 0$).

The market is cleared at the price of the last generator s (in the merit order) that supplies demand (and whose capacity is not binding). For generator s_1 this means $\overline{\mu_{s_1}^*} = 0 \Rightarrow o_{s_1} - \lambda^* = 0 \Rightarrow \lambda^* = o_{s_1}$.

Economic dispatch or one-node dispatch optimisation (III)

$$\min_{g_s} \sum_s o_s g_s$$

subject to:

$$\sum_s g_s - d = 0 \leftrightarrow \lambda$$

$$g_s \geq 0 \leftrightarrow \underline{\mu_s}$$

$$-g_s + G_s \geq 0 \leftrightarrow \overline{\mu_s}.$$

λ represents the change in the objective function at the optimal solution with respect to a small change in the constraint.

Small change in constraint:

$$d^* = d^* + 1 \text{ MWh}$$

Change in objective function:

$$\text{System cost}^* = \text{System cost} + \Delta\text{System cost}$$

λ represents the cost of supplying additional 1 MWh, i.e. the electricity price.

So far, this set of equations excludes the consideration of any network constraints (e.g. line limits), and any additional security constraints, and additional generation constraints (CO_2 emissions, ramp limits …)

Learning goals for this lecture

- Describe the optimal generator and demand behaviours and their relationship to maximising welfare.
- Explain how renewable and backup generators recover their capital and variable costs via market revenues.
- Interpret the meaning of Lagrange/KKT multipliers associated with the constraints in joint capacity and dispatch optimisation problems and analyse their values.
- Formulate a capacity and dispatch optimisation on a computer.

This lecture reviews the electricity market and shows why maximising profits for independent generators is equivalent to maximising social welfare.

For a comprehensive discussion and demonstration that the market-clearing problem obtains the Nash equilibrium solution (i.e. no market player desires to deviate from the market-clearing results), check the DTU course 46755 Renewables in Electricity Markets

Dispatch optimisation in one node

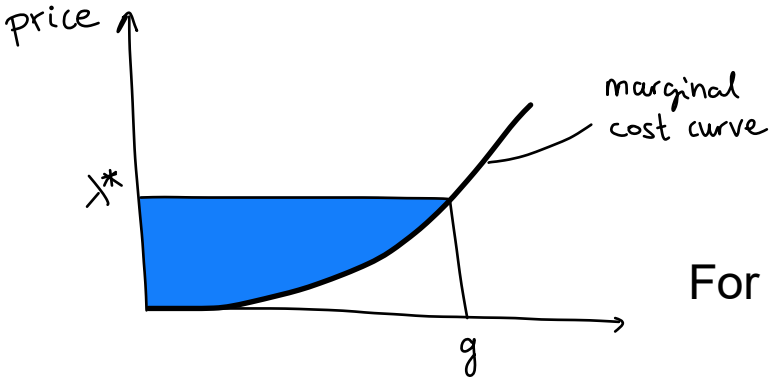
Optimal generator behaviour

If we assume that a generator is a price-taker (it cannot influence the market price λ) that wants to maximise its profits (revenues minus expenditures)

$$\max_g [\lambda g - C(g)]$$

where g is the energy generated and $C(g)$ the cost curve of the generator.

The generation g^* that maximises the generator's profits fulfils $\lambda - \left.\frac{\partial C}{\partial g}\right|_{g=g^*} = 0 \Rightarrow \lambda = \left.\frac{\partial C}{\partial g}\right|_{g=g^*}$



The marginal cost curve $\frac{\partial C}{\partial g}$ is the supply curve for a competitive firm. It shows for each generation g at which price the generator is willing to supply.

For a generator with constant marginal costs $C(g) = o_s \cdot g \Rightarrow \lambda = \left.\frac{\partial C}{\partial g}\right|_{g=g^*} = o_s.$

The maximum generation can be constrained, e.g. by the installed capacity

$g_s \leq G_s \leftrightarrow \overline{\mu_s}.$

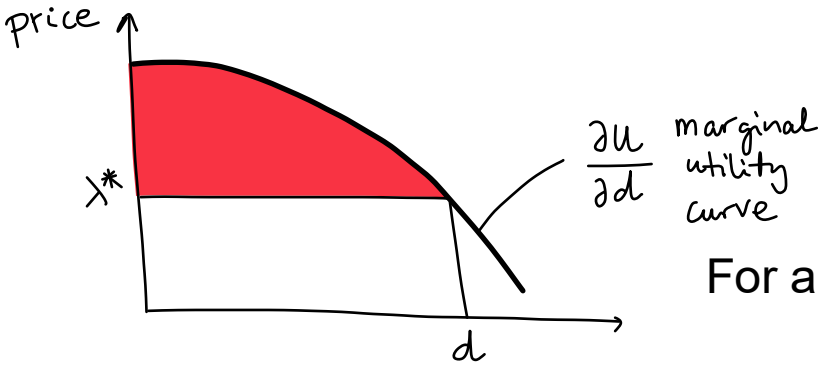
Optimal consumer behaviour

If we assume that a consumer is a price-taker (it cannot influence the market price λ) which wants to maximise its net utility or profits (utility minus expenditures)

$$\max_{d}[U(d) - \lambda d]$$

where d is the energy consumed and $\mathrm{U}(d)$ the utility curve of the consumer.

The consumption d^* that maximises the consumer's net utility $\left.\frac{\partial U}{\partial d}\right|_{d=d^*} - \lambda = 0 \Rightarrow \lambda = \left.\frac{\partial U}{\partial d}\right|_{d=d^*}$



The marginal utility curve $\frac{\partial U}{\partial d}$ is the demand curve for a rational consumer. It shows for each demand d at which price the consumer is willing to buy.

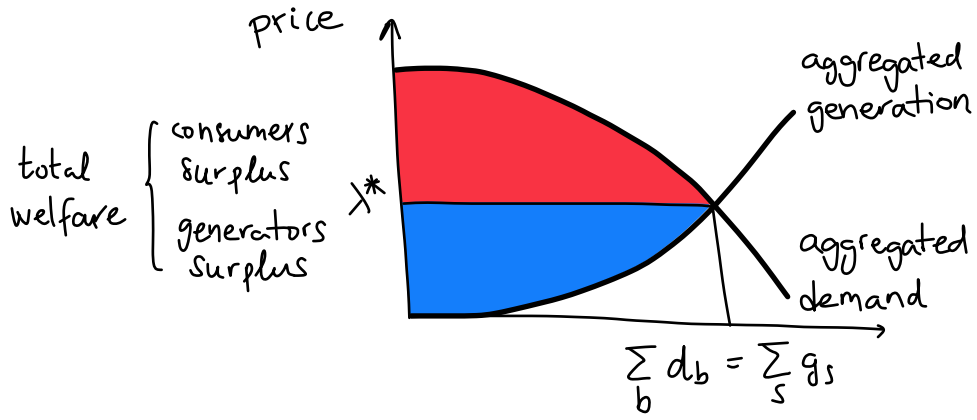
For a consumer with constant marginal utility $U(d) = o_b \cdot d_b \Rightarrow \lambda = \left.\frac{\partial U}{\partial d}\right|_{d=d^*} = o_b.$

The maximum demand can be constrained, e.g. by the capacity of the electrical machinery in the factory $d_b \leq D_b \leftrightarrow \overline{\mu_b}.$

Dispatch optimisation in one node

The total welfare (consumers and generator surplus) is maximised at the point where aggregated marginal cost (offer) and aggregated marginal utility (demand) meet

$$\begin{cases} \max\limits_{d_b, g_s} [\sum\limits_b U_b(d_b) - \sum\limits_s C_s(g_s)] \\ \text{subject to:} \\ \qquad \sum\limits_b d_b - \sum\limits_s g_s = 0 \leftrightarrow \lambda \end{cases}$$



$$\mathcal{L}(d_b, g_s, \lambda) = f(d_b, g_s) - \lambda h(d_b, g_s) = [\sum_b U_b(d_b) - \sum_s C_s(g_s)] - \lambda[\sum_b d_b - \sum_s g_s]$$

$$0 = \frac{\partial \mathcal{L}}{\partial d_b} = \frac{\partial f}{\partial d_b} - \lambda \frac{\partial h}{\partial d_b} = \frac{\partial U}{\partial d_b} - \lambda \qquad = 0 \Rightarrow \lambda^* \qquad = \left.\frac{\partial U_b}{\partial d_b}\right|_{b=b^*}$$

$$0 = \frac{\partial \mathcal{L}}{\partial g_s} = \frac{\partial f}{\partial g_s} - \lambda \frac{\partial h}{\partial g_s} = -\frac{\partial C}{\partial g_s} + \lambda \qquad = 0 \Rightarrow \lambda^* \qquad = \left.\frac{\partial C_s}{\partial g_s}\right|_{g=g^*}$$

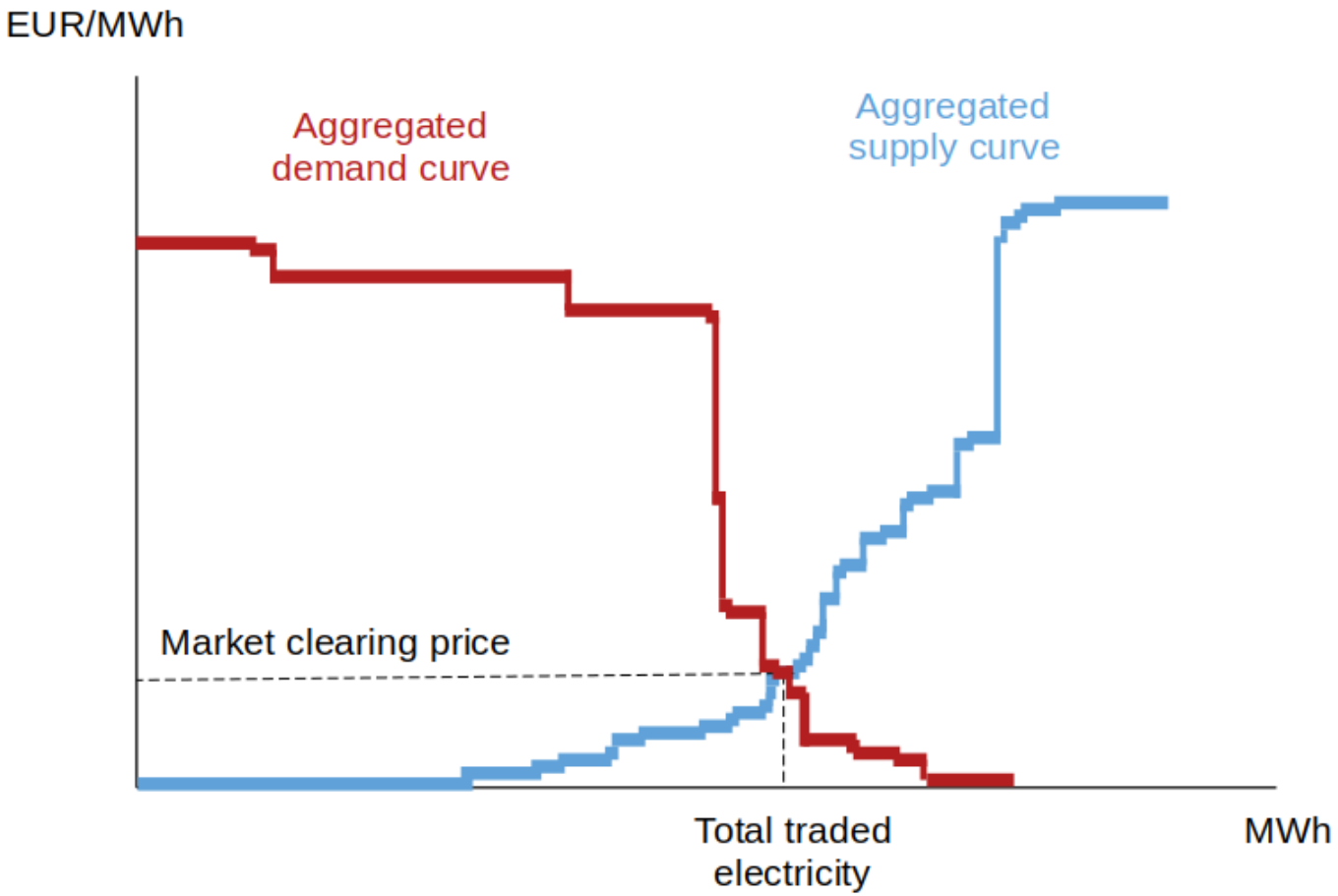
At the optimal point of maximum total economic welfare ("competitive equilibrium"), we get the same result as if everyone maximises their own welfare separately → The first welfare theorem in microeconomics justifies electricity markets.

Dispatch optimisation in one node – Electricity markets

The electricity market is the same as markets for coffee, wood, or any other goods

There are a few relevant differences:

- Supply and demand of electricity must be equal at any time step
- Demand is typically inelastic (in the short run)
- The resulting economic optimal solution must be compatible with power flow equations on the networks

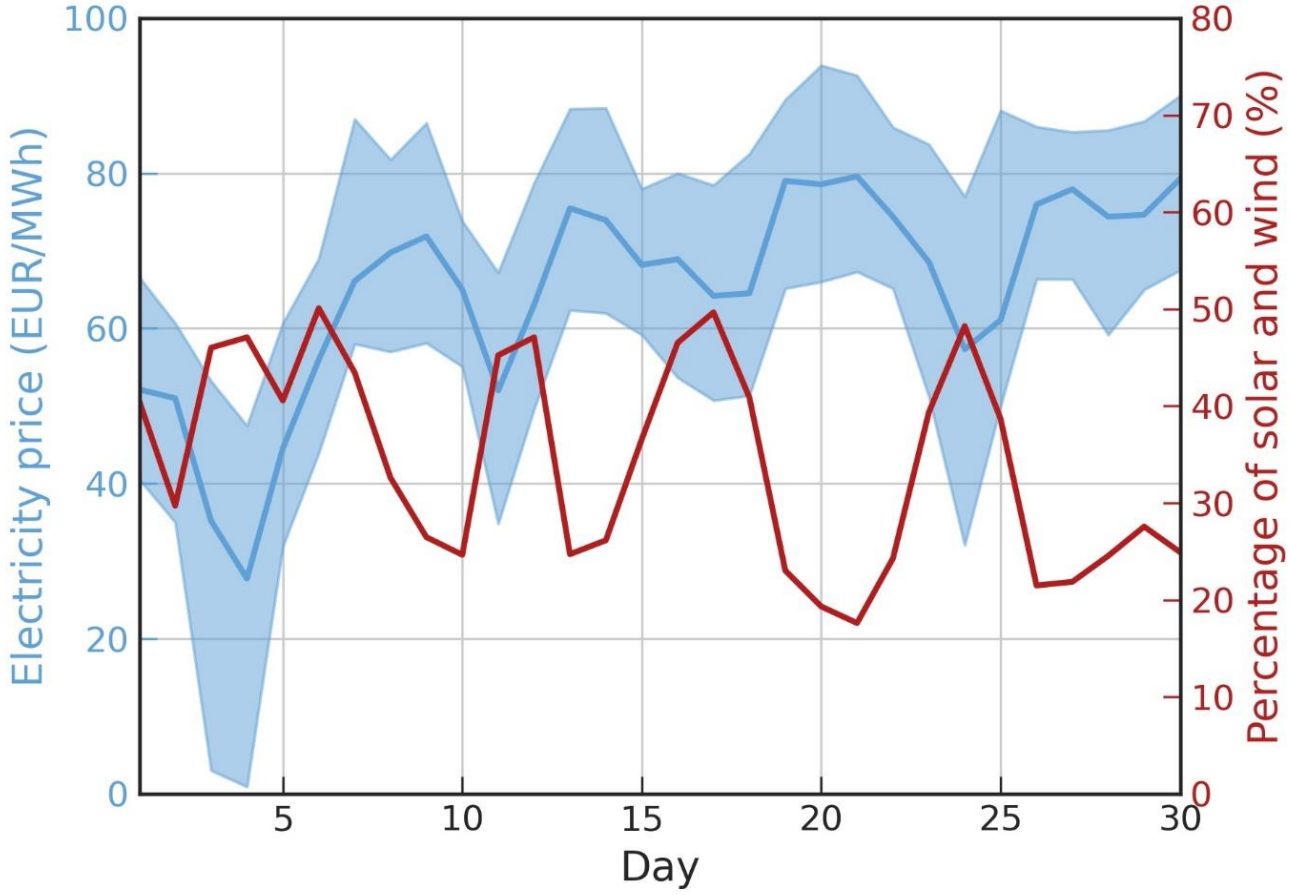


Ref: Victoria and Gallego, 2024, Economics of Photovoltaic Solar Energy (open-license figure)

For a comprehensive discussion on electricity markets, check the DTU course 46755 Renewables in Electricity Markets

Dispatch optimisation in one node – Electricity markets

Increasing penetration of renewable generation reduces the market price. This is known as the merit-order effect.



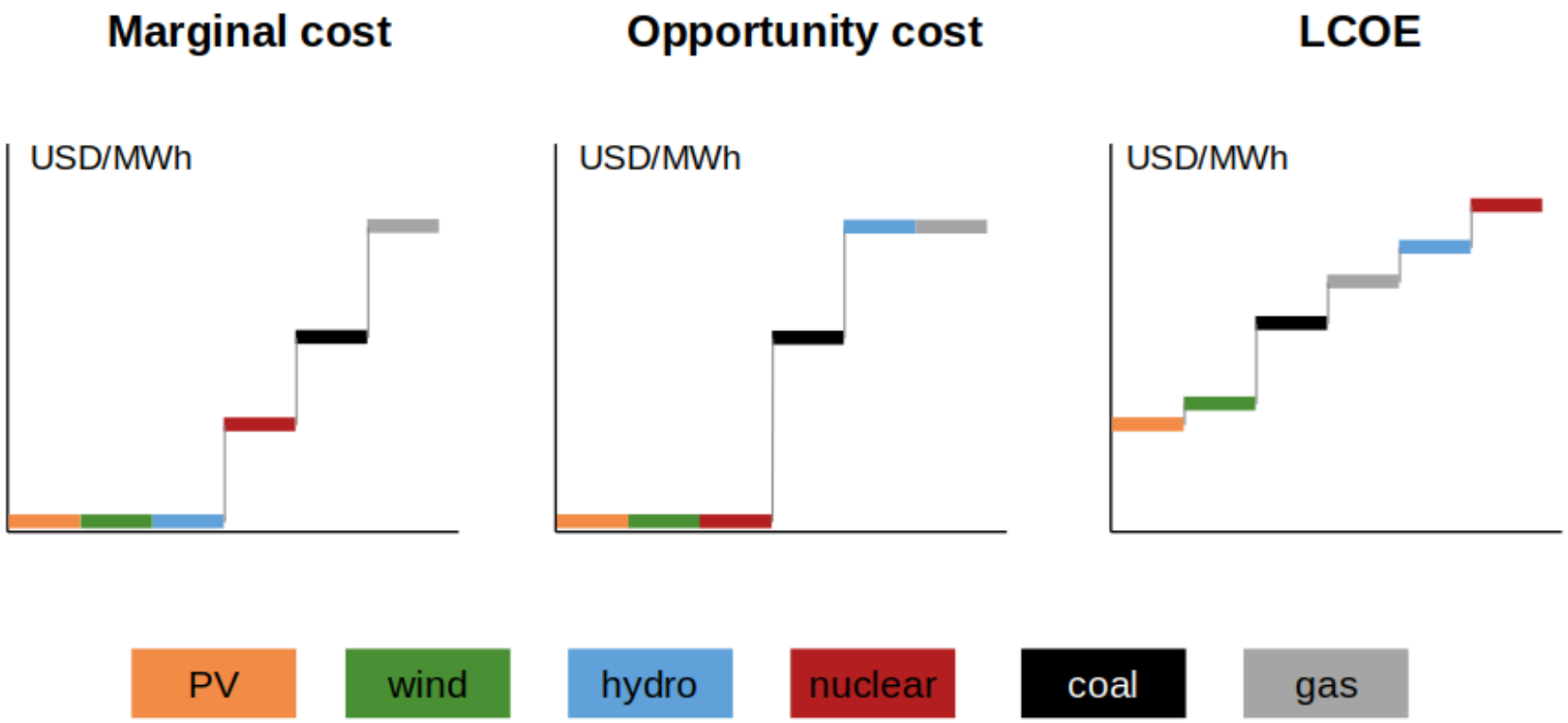
Ref: Victoria and Gallego, 2024, Economics of Photovoltaic Solar Energy (open-license figure)

Dispatch optimisation in one node – Electricity markets

Optimal generator behaviour is not only impacted by marginal generation cost but also by opportunity cost.

Opportunity cost = “loss of potential gain from other alternatives that could have been chosen”

Other alternatives = not producing (or producing at another time)



Ref: Victoria and Gallego, 2024, Economics of Photovoltaic Solar Energy (open-license figure)

Example: Opportunity costs in electricity markets

We are operating a hydropower reservoir. We have enough stored water to produce 100 MWh of electricity (dispatch at 100 MW for one hour). Our marginal cost of generation is 20 EUR/MWh. We cannot influence the electricity price with our decisions.

Noon

- Sunny (high solar potential), windy
- Medium demand
- Price forecast: 30 EUR/MWh

Decision: do we dispatch now?

- Revenue: 30 EUR/MWh
- Marginal cost: 20 EUR/MWh
- Profit: 30 – 20 EUR/MWh = **10 EUR/MWh**

Evening

- Dark (no solar), windy
- Demand: high
- Price forecast: 80 EUR/MWh

Decision: do we dispatch now?

- Revenue: 80 EUR/MWh
- Marginal cost: 20 EUR/MWh
- Profit: 80 – 20 EUR/MWh = **60 EUR/MWh**

We did not consider the opportunity costs!

- Opportunity cost: 60 EUR/MWh (evening profits)
- Net: 10 – 60 EUR/MWh = **-50 EUR/MWh**

- Opportunity cost: 10 EUR/MWh (noon profits)
- Net: 60 – 10 EUR/MWh = **50 EUR/MWh**

Maximising welfare

At the optimal point of maximum total economic welfare, we get the same result as if everyone maximises their own welfare separately → This is a fundamental results in microeconomics that justifies electricity markets.

Welfare can be maximised with decentralised markets:

1. If the market price is equal to the Lagrange/KKT multiplier of the supply-demand balance constraint.

2. All consumers are price-takers under perfect competition*.

* Perfect competition: price takers, homogeneous goods, perfect information, no entry/exit barriers, no externalities

Maximising welfare is equivalent to minimising total cost

For the sake of simplicity, we will **assume** (1) price-inelastic demand (i.e., it does note respond to price), (2) consumers with linear utility and (3) generators with linear costs

$$\max_{g_s}[o_d D - \sum_s o_s g_s]$$

subject to:

$$D - \sum_s g_s = 0 \leftrightarrow \lambda$$

In this simple case, since D is a constant, maximising welfare is equivalent to minimising aggregated cost.

Maximising welfare is equivalent to minimising total cost

We can assume now elastic demand d and represent demand elasticity or load shedding by adding a dummy generator g_d so that $d = D - g_d$.

$$\begin{cases} \max\limits_{g_s} [o_d D - o_d g_d - \sum\limits_s o_s g_s] \\ \text{subject to:} \\ \qquad D - \sum\limits_s g_s = 0 \leftrightarrow \lambda \end{cases}$$

We can group the dummy generator with the rest of generators and again we can see that maximising welfare is equivalent to minimising aggregated cost.

Joint capacity and dispatch optimisation in one node

Joint capacity and dispatch optimisation in one node

Now our optimisation variables are the energy $g_{s,t}$ generated by every generator s in every **time step** t and the installed capacity G_s of every generator

$$\min_{g_{s,t},G_s} \left[\sum_s c_s G_s + \sum_{s,t} o_s g_{s,t}\right]$$

subject to:

$$\sum_s g_{s,t} - d_t = 0 \leftrightarrow \lambda_t$$

Energy balance in every time step

$$-g_{s,t} + G_s \geq 0 \leftrightarrow \overline{\mu_{s,t}}$$

Generation limited by installed capacity in every generator and time step

$$0 = \frac{\partial \mathcal{L}}{\partial g_{s,t}} = \frac{\partial f}{\partial g_{s,t}} - \sum_i \lambda_i \frac{\partial h_i}{\partial g_{s,t}} - \sum_j \mu_j \frac{\partial g_j}{\partial g_{s,t}} = o_s - \lambda_t^* + \overline{\mu_{s,t}^*} = 0 \Rightarrow \overline{\mu_{s,t}^*} = \lambda_t^* - o_s.$$

For given capacities, the optimal hourly dispatch is the same as the single period economic dispatch → For every time step t, the generator needed so that the supply curve intersects the demand: $\lambda_t^* = \left.\frac{\partial C_s}{\partial g_s}\right|_{g=g^*} = o_s.$

Joint capacity and dispatch optimisation in one node

We can also derive the Lagrangian with respect to the installed capacity G_s of every generator

$$\min_{g_{s,t},G_s} \left[\sum_s c_s G_s + \sum_{s,t} o_s g_{s,t}\right]$$

subject to:

$$\sum_s g_{s,t} - d_t = 0 \leftrightarrow \lambda_t$$

$$-g_{s,t} + G_s \geq 0 \leftrightarrow \overline{\mu_{s,t}}$$

From previous slide

$$0 = \frac{\partial \mathcal{L}}{\partial G_{s,t}} = \frac{\partial f}{\partial G_{s,t}} - \sum_i \lambda_i \frac{\partial h_i}{\partial G_{s,t}} - \sum_j \mu_j \frac{\partial g_j}{\partial G_{s,t}} = c_s - \sum_t \overline{\mu^*_{s,t}} \cdot (1) = 0 \Rightarrow c_s = \sum_t \overline{\mu^*_{s,t}} = \sum_t (\lambda^*_{s,t} - o_s).$$

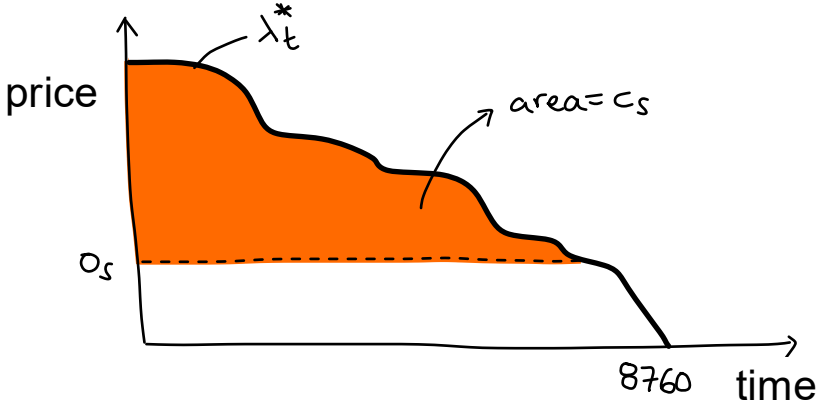
The level of investment in a generator's capacity is optimal when the sum of the gap between the electricity price λ and the generator's marginal cost o_s is equal to the capital cost of the added generation capacity.

Joint capacity and dispatch optimisation in one node

$$c_s = \sum_t \overline{\mu^*_{s,t}} = \sum_t (\lambda^*_{s,t} - o_s)$$

From previous slides

The level of investment in a generator's capacity is optimal when the sum of the gap between the electricity price λ and the generator's marginal cost o_s is equal to the capital cost of the added generation capacity.



The price of the electricity market must be higher than the variable cost for the most expensive generator for several time steps. In those time steps, the expensive generator can cover its capital cost.

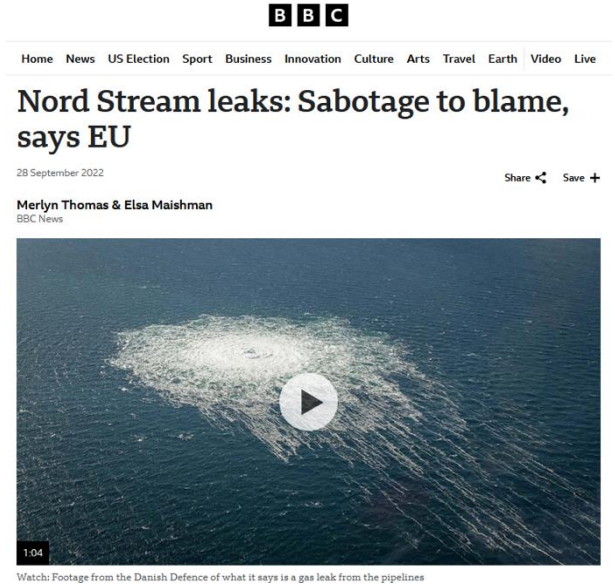
$$c_s = \sum_t \max(0, \lambda^*_t - o_s)$$

(strictly speaking, if we do not generate, the dual variable satisfies $\overline{\mu^*_{s,t}} = 0$)

High electricity prices in energy markets

High electricity prices:

- Are a feature, not a bug: ensure that the most expensive generators recover their capital costs
- Create concerns for consumers and whole society
- Are a sign of scarcity (renewable droughts, gas scarcity, …)
- Are a consequence of inelastic demand (and could be smoothed by demand elasticity and additional electricity demand from other sectors)
- Can indicate market power (assumption of perfect competition is not valid anymore)
- How do we pay for (peaking) capacities that ensure resilience but are rarely used throughout the year?



This is a hot discussion topic for European energy markets, some interesting angles:

- Fabra 2018, A primer on capacity mechanisms, Energy Economics
- Brown et al. 2025, Price formation without fuel costs https://doi.org/10.1016/j.eneco.2025.108483

Problems for this lecture

Problems 3.1 a), b) (**Group 6**)

Problems 3.1 c), d) (**Group 7**)

Problems 3.1 e), 3.2 (**Group 8**)

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